

Sol

cal

Co1 AT-2

Total Spectrum = 45 MHz

Simple channel = 2.5KHZ

Duplex channel = 2 simplex

channels

No. of channels = $\frac{45000 \text{ KHZ}}{2.5 \text{ KHZ}} = 1800$

No. of FC = $1800 \div 2 = 900$

2 Given

1

Total Voice channels : 560

الون

cluster size = 7

bovignes

cal

boldwob quotzia

22 al-At

channels per cell

= $560 \div 7 = 80$

3

Given

Cal

Total

No. of carriers = 100

ان

Time slot per carrier = 8

boldweb of author 1195

الله

voic channels = $100 \times 8 = 80092$

Given

BSP = 80 W

bbq

нраиш

6947

cal

Power Ratio (dB) = $10 \log_{10}$

$\log_{10} (P1 / P2)$

given

CS=-60dBm

inter ference :-92dBm

21: Jon og algo

Cal

od

$$-60 - (-92) = 32\text{dB}$$

21 01 8 64gh

लक्ष

to be?

加

6 Given

Given

FA

Total spectrum=42MHz

channel bandwidth

= 30 KHz

cluster
size=7

Total channels
 $42,000 \div 300 = 140$

Los

0081 07 30 31

channels per cell
 $140 \div 7 = 20$

c

initial received power = buwis

rahulb Distance doubled

Path-loss exponent = 4

08

assade an rag alangodo

$$P_2 = P_1 (d_1/d_2)^2$$

Formula:

Given

Formula:

P_2

64

64

$=4w$

24

16

وی آن جو

cell radius is doubled

Inverse **square** law

($n=2$) **jou**

Loron

Pad 2

was

2

$P_{\text{new}} = 2 = 4$

Given

oi: (ab) othe

9 wal

cluster size : 4

channels per cell: 15

body 2

cal

$$4 \times 15 = 60$$

10 Given

$$\text{speed} = 100 \text{ km/h} = 27.78 \text{ m/s}$$

$$\text{Frequency} = 900 \text{ MHz}$$

Speed of light

$$= 3 \times 10^8 \text{ m/s}$$

"

Formula:

$$fd = ve$$

cal

Given :

cal

12

Given

F

بنا

ارد

6

194

جماله

$$fd = 27.78 \times 900 \times 10$$

$$3 \times 10^8 \\ = 83.3 \text{ Hz})$$

TDMA frame duration = 5ms

Number of time slots = 8

02/10004

on,ber: 02 SOOON

Slot Duration : 5 : 0.625 m s

50

Total bandwidth = 24MHZ brod

Jonmodo

carrier spacing=200KHZ

204021

cal

241000 KHZ

= 120

200KHZ

brown

Given:

Wb7od

power post tagoj ba

Transmit power =8w

Path

PARODY-ONX (108)

loss = 33 dB

step1 : -

step 2:-

$P_{dBm} = 10 \log 10$

$(8000) = 39-03dBm$

$$P_r = 39.03 - 33 = 6.03 \text{ dBm}$$

دولا دانش

cuven

cal

Frequency = 1900 MHz

OP

Distance: 2.5km

ballov

ap

ما

D

$$(2.5) + 20 \log_{10} (19000)$$

$$= 32.44 + 7.96 + 65.58 = 105.98 \text{ dB}$$

Given

Number of cells = 6

calls per
cell = 250

cal
 $6 \times 250 = 1500$

16
Given .

Cal

(1)
Given

caf

18

Given

call requests: 4000 calls (sec

Average call duration : 3 min = 180 sec

2438

$$4000 \times 180 = 720,000$$

channel 150

channel bandwidth = 25ktyabred Jotor

$$150 \times 25 = 3750 \text{ KHZ} = 3.75 \text{ MHZ}$$

Guard band = 40KHZ

100

Adjacent frequency

bands = 30

Cal

YSWOT Hasdorf

$$(30-1) \times 40 = 1160\text{kHz}$$

(0888-3

3 90

= 1.16 MHz

(a) aiven : -

Speed = 90km/h

88

cell radius = 1.5km

cal

20

Given

Regional cente

Cal

Base station

UPBS = 250

$8 \times 40 = 320$

$320 \times 250 = 80$

(2)

21

Given:-

Total area =

Caf

cell radius=1

Approximate

$CA = (0-4)^2 =$

No. of cell-

300

0.16

22

Given.

Jo

Total specfurm

Channel

band

Cal

در

6000 KHZ 30KHz

1992

23 given

Jo1 maha

Subscribers

cell diameter = $2 \times 1.5 = 3\text{km}$

Намом д

Distance travelled in Thour = 90km

90

3
19

ترلر
4421

30. pp 8 =

Cal

cell per

heur Avero

20 Given

Cal

1200 хих

Total bar

Bandwidth

401

20

Given

cal

Regional centers = 8

Base station per center = 40

UPBS = 250

8 x 40-320

320 x 250 = 80,000

eco Der-cod Xoo

24 Given:

Total

area = 400km?

cell radius = 400m=0.4km

ee

062

Approximate cell area = R^2

$$CA = (0.4)^2 = 0.16 \text{ km}^2$$

000

2500

تمار

درا

0.163.000 jostavaltio 100

No. of cell = 400

22 Given .

dobroest

داری

Total specfurm=6MHY

000,05) =5x000101°

channel

Cal

bandwidth:30 KHZ 0010001001

Given

6000 KHZ

=200

30KHy

000,000

000,00

G00,00

Cal

20 Given

Subscribers = 1200

cell

CHA

Per subscriber: ucalls / hour

Average cell duration - 2.5
min

1200 хих

Total

х
2.5

60
= 1200 × 0.1667=200

goo

bandwidth = 40MHZ

600
0000

Bandwidth

Cal

per
aeser = 400 KH Z

2007

40,000KHZ

= 100

400KHZ

0018

0.0001

7

تھا

R

25) Given

BS= 1200

User per bs=600

022-839 0

Cal

$1200 \times 600 = 720,000$

26

alven

Total cell 200

cell per round = 4

cal

200

=50

! دول

mugs moon wibor los @go to
plowitangga

34

27

Given

Network capacity = 100,000

031

call attempts = 40,000/min ag

Cal

Average call duration 3 minutes

Ma

120,000 - 100,000 20,000 bind Jodant's

40,000 × 3 = 120,000 lptor

32 Giv

20,000

120,000

$\times 100 = 16.67\%$

11 good

cal

008

Given

Total subscribe: 600,000

Roaming

User =15%

ni

BUON 200

dinsdua nga

دول

193

600, 000X.

15

water 0,000

100

gorau Hom

29

Given

SCC = 12,000

current load

load = 9,000

HMON=Arbicubreed

yif ody

9.03

Arbicubras

9000

12000

X100=75%.

H000,0

INYOOM

دراسی

33

Given

ليه

34 Given:

CS =

User

30 Given

TS = 48MHZ

CS = 200 KHZ

F requvey factor = 4

Step 1:.

48,000

= 240

200

Step 2:-

240

= 60

U

U

34

Given

To find

specfurm = 30MHZ

$C_S = 200\text{KHz}$

لو

• Fal? To

207 da

نام

67

Number =

of channels.

30,000 KHZ

200 KHZ

=50

32 Given

Total

Success rate=99%. ۹۹.

cal

handoff attempts =

250000020-514

000,05 – Jouraco 100

- هان

Given

2500x 900

=2475

100

TP=12W

PL=5dB

000108-000 03

0.00.08

ood of

00) x

000,0% 9

Caf

داسان 2

PdBm = 10 109 10 (12000)

$$= 40.79 \text{ dBm}$$

$$X=40$$

$$P_Y = 40.79 - 45 = -4.21 \text{ dBm}$$

2930

112

34 Given:

$$cs = 7 \text{ cells}$$

ood xp

User cell: 900

$$7 \times 900 = 6300$$

35

Given

$$\text{TDMA frame} = 4.8 \text{ ms}$$

Number

$$\text{of slots} = 8$$

Ca

$$\frac{9.8}{36} = 0.6 \text{ms}$$

36
Given
 ∞
شروط

$$\text{Speed} = 150 \text{km/h} = 41.67 \text{m/s}$$

1414

ONS

تر

Frequency = 1.8GHz = 1.8×10^9 Hz

Speed

of

light = 3×10^8 m/s

3×10^8 998

formula

$f \lambda = v$

$v = f \lambda$

Joll

37

Given

این

38

Given

cal

000.

$$fd = 41.67 \times 1.8 \times 10$$

$$3 \times 10^8$$

$$= 250 \text{ Hz}$$

$$NC = 250,000 \times 2$$

$$\text{Call arrival} = 70,000$$

$$\text{Average call} = u_{\min}$$

$$10,000 \times 4 = 280,000$$

$$2510$$

$$J0J$$

$$280,000 - 280,000 = 30,000$$

$$30,000$$

$\times 100 = 10.71\%$.

280,000

cluster size = a cell boost) polo)

—

user

orab

per cell = 600

9x600-5400

Jolet

روائع

21955-29

(39) Given

او

call process - 60,000

Nim soboth th

Doop rate = 0.4%

$$60,000 \times \underline{0.4} = 240$$

toO

۹۰

Given

TP=43dBm

nabas

Longiz

Received power = -87 dBm

-87dBm 190009

formilla

Path Loss = $P_t - P_r$

abos- (angas-

cal

$$43 - (-87) = 130 \text{ dB}$$

20 to on

(a) Given

Area

29840213 920 rog arbicklund

covered by one base station

$$= 2.5 \text{ km}^2$$

2944008

No. of base stations = 1500028

293

stations = 4500028

al

2

$$150 \times 2.5 = 375 \text{ km}$$

12

Given

Caf

TS=48MHZ

CB=200KHZ

F F = b

028

48,000KHz

9021298?

radiadwe

002!

od

= 240

200KHZ

alamoto Joa

240

= 40

So rag dibice brag

6

Cuven

user

process per cyclo = 6000

TS = 150,000

Caf

150,000

= 25

61000

0908

(44) Given

Cal

Call duration = amin

HE 3 min

45

Given

Cal

$9=3$

3

120

Signal Power: -85 dBm

Noise power = -115 dBm

$-85 - (-115) = 30\text{ dB}$

229) dhe

46 Given

No. of User = 250

Bandwidth per

User: 150 kbps

Caf

$$250 \times 130 = 37,500 \text{ kbps}$$

$$37,500 \text{ Kbps} = 37.5 \text{ Mbps}$$

$$\text{Subscribe} = 1500$$

47 Given

call per

JA

Cal

$$\text{Average call} = 2.5 \text{ min}$$

subscriber I u calls / nour

$$1500 \times 4 \times 2.5 = 1500 \times 0.1667 =$$

250

$$1500 \times 4 \times 2.5$$

60

48 Given

cal

control channels = 250

Band width per channel 212kHz

$$200 \times 12 = 3000 \text{ KHZ}$$

$$3000 \text{ KHZ} = 3 \text{ MHz}$$

How 30-27

H

49 Given

Simulato Data rate

good
cal

60 Given
1800×
900!

COU
cal
civen

Simultaneous = 1500
Data rate
per connection = 600kbps
cal

60 **Given**

$$1500 \times 600 = 900,000 \text{ kbps}$$

$$900,000 \text{ Kbps} = 900 \text{ Mbps.}$$

Tower coverage radius

$$= 4 \text{ km}$$

coverage

area

per tower.

cat

2

$$A = 3 \cdot 14 \times 42 = 3 \cdot 14 \times 16 \cdot 50 \cdot 24$$

km²

350

50.24

26.97